

Reflection Journal – Module 2: ML Development Tools

This lab helped me understand some of the basic tools that are used in machine learning, but it also showed me which areas I still need to practice. The most challenging parts for me were understanding NumPy and interpreting the graphs created with Matplotlib. Before this lab, I knew that NumPy was connected to mathematical operations in Python, but I did not clearly understand why it was useful when working with data. During the exercise, I saw NumPy being used to calculate the mean and standard deviation of the sepal length values in the Iris dataset. This helped me understand that NumPy can perform mathematical calculations on collections of numbers quickly and efficiently. I still need more practice with NumPy, especially understanding arrays and knowing when I should use NumPy instead of other tools such as pandas.

Working with graphs was another important learning experience for me. At first, looking at the code used to create a graph with Matplotlib was confusing because there were several instructions controlling the size, labels, colors, title, and data being displayed. Once I saw the completed scatter plot, however, I started understanding why visualization is useful. Instead of only looking at rows and numbers, the graph allowed me to see relationships between sepal length and sepal width for the different Iris species. This helped me realize that graphs are not just for making data look attractive. They can help a person recognize patterns and differences in a dataset that may be harder to notice when looking only at numerical values.

Another thing I learned is how different machine learning tools work together. Pandas made it easier to organize the Iris dataset into a DataFrame and examine the information. NumPy was useful for numerical calculations, while Matplotlib transformed the data into visual information. Before this lab, I tended to think of these libraries as separate tools. Now I understand that they can be combined as part of the same data analysis workflow. Each tool has a different purpose, but together they make it easier to understand and prepare data.

This lab also changed the way I think about learning Python for machine learning. I realized that I do not need to understand every line of code immediately. Running the cells and examining the output helped me connect the code with what it was doing. When I saw the mean, standard deviation, scatter plot, and bar chart appear after running the code, the concepts started becoming more understandable. I still have questions about how NumPy performs operations on large datasets and how to choose the best type of graph for different kinds of data.

Overall, this lab gave me a better understanding of the basic machine learning development environment and showed me areas where I need additional practice. NumPy and data visualization are still challenging for me, but they make more sense now than they did at the beginning of the lab. Going forward, I want to practice these tools more so that I can understand not only how to run the code, but also why each tool is being used and what the results mean.

